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MILITARY SPECIFICATION

INDICATOR, TURN AND SLIP, 2-INCH SIZE, INTEGRALLY LIGHTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for integrally lighted, 2-inch size, turn and slip indicators.

1.2 Classification. Indicators shall be of the following types and classes, as specified (see 6.2):

ARU-3/A	Integrally red lighted
ARU-19/A	Integrally white lighted
Class 1	Dark indicator ball
Class 2	White indicator ball

2. APPLICABALE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Federal

BB-N-411 - Nitrogen, Technical
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-636 - Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-P-116	-	Preservation, Methods of
MIL-C-5015	-	Connector, Electrical, Circular Threaded, AN Type, General Specification for
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-I-6181	-	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25467	-	Lighting, Integral, Red, Aircraft Instrument General Specification for
MIL-L-27160	-	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	-	Colors

Military

DOD-STD-100	-	Engineering Drawing Practices
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-471	-	Maintainability Demonstration
MIL-STD-781	-	Reliability Tests Exponential Distribution
MIL-STD-810	-	Environmental Test Methods
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-838	-	Lubrication of Military Equipment
MIL-STD-889	-	Dissimiliar Metals
MS 33558	-	Numeral and Letter, Aircraft Instrument Dial, Standard Form of
MS 33678	-	Connector, Receptacle, Electric, Integral Mounting

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets and MS standards), the text of this specification shall supersede applicable laws and regulations unless a specific exemption has been obtained.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specification acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.2).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray or atmospheric conditions likely to be met in storage or normal service.

3.3.5 Dissimiliar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5.1 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3.6 Corrosive fumes. The materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.7 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The indicator shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Adjustments and repairs. The indicator shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases with tools normally available commercially.

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3.4.2 Maintainability. The following maintainability requirements shall be considered in the design of the indicator:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, etc.) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c. Rapid and positive identification of the replaceable defective part, assembly or component.

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

* 3.4.3 Reliability. The indicator shall have a lower test mean-time-between-failure (MTBF) of 2000 hours at a confidence level of 90 percent and a discrimination ration of 1.5:1.

* 3.4.3.1 Longevity. The indicator shall have an operating life span of not less than 3000 hours before a decrease in MTBF occurs. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

3.4.4 Electrical receptacle. The electrical receptacle, provided as a part of the indicator, shall be an MS 33678-R14-2P sealed receptacle conforming to MIL-C-5015.

3.4.4.1 Markings. The following markings shall be in durable dull white or natural metal finish and shall be permanently and legibly marked on the back of the case beside the electrical receptacle to identify distinctly the electrical receptacle pins to which they apply:

Pin	Connection
A	+28V dc
B	-28V dc
C	5V (lighting)
D	Lighting ground

3.4.5 Power supply. The indicator shall be designed to operate from a 28V dc nominal power supply. The integral lighting shall operate from a 5V dc nominal power supply.

3.4.6. Cover glass. The cover glass shall be clear, flat, and free from flaws that might interfere with reading of the indicator. When exposed to ultraviolet light or rays in the 365-millimicron region, the cover glass shall not be fluorescent.

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3.4.6.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be held to a minimum and shall not exceed 0.025 inch.

3.4.7 Dial. The dial shall conform to figure 1. All markings shall be so durable as to withstand usage encountered in service. The style and proportions of the numerals and letters on the dial shall conform of MS 33558.

3.4.8 Motor. The design of the indicator motor shall not include a commutator, slip rings, contacts, or any similar element that might fail due to wear incurred in normal operation.

3.4.9 Slip indicator.

3.4.9.1 Tube. The slip indicator tubing shall be made of clear, annealed glass which shall be free of any flaws that would affect the readability of the inclinometer. The wall shall not be less than 3/64 inch. The inside of the tube shall be so smooth and uniform that the ball will roll smoothly.

3.4.9.2 Ball. Class 1 indicator balls shall be black glass, color no. 17038, or dark blue glass, color no. 15044, conforming to FED-STD-595. Class 2 indicators shall be white glass in accordance with the following Munsell system color notation: Neutral not less than 9.0 value nor greater than 0.5 chrome. Material other than glass may be used only when approved by the procuring activity. The ball shall be a highly polished, accurate sphere which shall have a diameter of 3/8 $\pm$ 1/64 inch.

3.4.9.3 Damping liquid. The damping liquid in the slip indicators shall be sufficiently colorless to preclude interference with satisfactory silhouette illumination of the ball by luminescent material.

3.4.10 Turn indicator.

3.4.10.1 Sensitive element. The sensitive element of the turn indicator shall be operated on 28V dc. The element shall be so mounted that it will react only to turns about the veritcal axis; the indication shall be proportional to the rate of turn about the vertical axis.

3.4.10.2 Pointer. The pointer shall conform to figure 1. It shall be as light as practicable, sufficiently rugged to prevent oscillation under vibration, and shall be firmly attached to the mechanism and still be readily adjustable.

3.4.10.3 Adjustment mechanism. Means shall be provided for adjusting or correcting the sensitivity and, if necessary, the damping of the indications. The adjustment mechanism shall be set in the neutral position, within one-quarter of the total adjustment range, so that the adjustment range in each direction will be sufficient. The adjustment mechanism shall be simple to manipulate and so designed that adjustments can be accomplished with tools ordinarily used by instrument repairmen.

3.4.11 Integral lighting. Integral lighting for the ARU-3/A indicator shall conform to MIL-L-25467, and lighting for the ARU-19/A shall conform to MIL-L-27160.

3.4.11.1 Prime standard. The prime standard shall be either an ARU-3/A or

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ARU-19/A turn and slip indicator, as applicable, conforming to this specification or a lighted model. All production units shall be visually compared to the prime standard for approval and acceptance. The internal lighting of the prime standard shall be subject to the approval of the contracting activity.

3.4.11.2 Average areas. A minimum of 12 average areas covering all representative areas of the face of the prime standard indicator shall be measured periodically (at least every 50 hours of operation) for brightness and color to insure proper calibration of the prime standard.

* 3.5 Performance. The indicator shall operate as specified herein under any combination of the following conditions:

a. Temperature - ambient operating temperatures ranging from -62° to $+71^{\circ}$C.

b. Humidity - relative humidity up to 100 percent.

c. Acceleration - acceleration forces up to 20g's.

d. Altitude - pressures ranging from 30 inches Hg (sea level) down to 0.082 inch Hg (approximately 80,000 feet).

e. Fungus - exposure to fungus enzymes.

f. Salt fog - exposure to salt fog for a minimum period of 168 hours.

g. Dust - exposure to blowing fine sand and dust particles.

h. Temperature shock - temperature shock caused by abrupt temperature drop from +85 to -54°C.

* 3.5.1 Audio frequency susceptibility. The indicator shall meet the audio frequency susceptibility requirements specified in MIL-I-6181.

3.5.2 Static balance. With no power applied, pointer deviation from zero shall not exceed 0.010 inch for any stationary position.

3.5.3 Dynamic balance. With 28V dc applied, pointer deviation from zero shall not exceed 0.010 inch for any stationary position.

3.5.4 Dielectric strength. The indicator shall withstand a potential of 500V dc applied between the positive, negative, and conducting parts of the case, each for a period of 60 seconds.

3.5.5 Turn sensitivity. Turn sensitivity shall be within the specified limits when the indicator is operated for a period of not more than 3 minutes.

3.5.6 Slip sensitivity. When the indicator is rotated through an angle of $8^{\circ} \pm 2^{\circ}$, the ball shall not stick at the high end of the tube.

3.5.7 Leakage. The leak rate in the case shall not exceed 0.1 micron cubic foot per hour.

3.5.8 Damping. While being operated and rotated about the vertical axis to full-scale deflection, the pointer shall return to zero with no overswing.

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3.5.9 Vibration error. When subjected to amplitudes ranging from 0.003 to 0.005 at frequencies from 5 to 50 Hz, the turn indicator shall not oscillate more than 1/32 inch and the slip indicator shall not oscillate more than 1/16 inch.

3.5.10 Vibration failure. The indicator shall withstand vibrations up to 5g's at frequencies ranging from 5 to 500 Hz.

3.5.11 Voltage extremes. The indicator shall meet the sensitivity requirements specified herein when 26 and 30V dc are applied to the indicator terminals.

* 3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Case. The case shall conform to figure 2. It shall be hermetically sealed, made of sturdy material which shall be uniform in texture, and shall be finished in durable dull black. The hermetic seal shall not be accomplished by use of materials that will be affected by the action of any atmosphere to which the case may be subjected. The case shall be so designed that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the indicator.

3.7.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligrams of water vapor per liter (dewpoint $-65^{\circ}C$) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. The nitrogen shall be in accordance with BB-N-411, type I, grade A.

3.8 Markings.

3.8.1 White. The following marking shall be durable dull white in accordance with FED-STD-595, color no. 37875. The dimensions shall be as shown on figure 1.

Lettering 4 MIN TURN
Two needle width deflection mark
Shaded portion of pointer
Rear half and bottom of
slip indicator tube

3.8.2 Black. The following markings and dimensions shall be durable dull black in accordance with FED-STD-595, color no. 37038.

Height or length inch	Width of line or graduation inch
± 0.010	± 0.005

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Visible wires each side of inclinometer ball	—	0.031
ARU-3/A or ARU-19/A	0.047	—
Lettering TURN AND SLIP INDICATOR	0.047	—
Dial background	—	—

3.9 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.10 Weight. The total weight of the completely assembled indicator shall not exceed 1.50 pounds.

3.11 Lubrication. When lubrication is required, it shall be selected in accordance with the procedures specified in MIL-STD-838 and shall be of such quality that proper lubrication will be provided for indicator operations at temperatures ranging from -54° to $+71^{\circ}$C. Each indicator shall be furnished with a sufficient quantity of lubrication for at least 1,500 hours of operation.

3.12 Finishes and protective coatings.

3.12.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.12.1.1 Exceptions. Dials, small holes, and case inserts need not be anodized.

3.12.1.2 Aluminum alloys. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.12.1.3 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.12.1.4 Castings. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film conforming to MIL-C-5541 in lieu of anodizing.

3.12.1.5 Abrasion. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Mounting screws. The contractor shall provide sufficient .138-32 UNC 2A roundhead brass machine screws of a length required for mounting the indicator on a panel 3/16 inch thick. The screws shall have a durable dull black oxidized or black nickel finish. They shall be placed in an envelope marked with the following information and packed with each instrument:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

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3.15 Workmanship. The indicator, including all parts and assemblies, shall be finished in a workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

3.15.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability or operation of the indicator, they shall be held or limited accordingly.

3.15.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage to the screw, bolt or threads.

3.15.3 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.15.4 Cleaning. The indicator shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other material removed after final assembly. Burrs and sharp edges as well as resin flush that might crumble shall also be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection and testing of indicators shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in paragraph 4.3.1, 4.3.2, 4.3.3 and 4.3.4.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test will be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with the atmospheric pressure at room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditon.

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4.3.2 Tapping. Unless otherwise specified, the indicator shall be lightly tapped before a test reading is taken.

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position.

* 4.3.4 Vibration. A vibration stand shall be used that will vibrate at any desired frequency between 5 and 50 Hz and will subject the indicator to vibration in such a manner that a point on the case will describe, in a plane inclined 45° to the horizontal plane, a circle having a diameter equal to the amplitude specified herein.

* 4.4 First article inspection.

* 4.4.1 Test samples. The test samples shall consist of 12 indicators representative of the production equipment. Three of the indicators shall be subjected to all the tests except reliability and the other nine indicators shall be subjected to the reliability test only.

* 4.4.1.1 Reliability demonstration. Unless otherwise specified, 9 indicators representative of the production equipment shall be tested as specified in 4.6.32. Accept/reject criteria shall be in accordance with MIL-STD-781C, test plan IC, except that no more than 10 failures will be allowed as shown on figure 3.

* 4.4.2 First article inspection. The first article inspection shall consist of all the tests described under paragraph 4.6.

4.5 Quality conformancce inspection. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Turn indicator static balance

c. Turn indicator dynamic balance

d. Dielectric strength

e. Turn indicator sensitivity

f. Slip indicator sensitivity

g. Slip indicator zero position

h. Slip indicator friction

i. Slip indicator visibility

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j. Slip indicator filling

k. Leakage

l. Lighting

m. Fogging

4.5.2 Quality conformance inspection sampling.

4.5.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced and subjected to the following tests, as described in paragraph 4.6:

a. Individual tests

b. Turn indicator damping at room temperature

c. Slip indicator damping at room temperature

d. Turn indicator sensitivity at low temperature

e. Slip indicator damping at low temperature

f. Turn indicator sensitivity at high temperature

g. Slip indicator leakage

h. Magnetic effect

i. Vibration error

j. Current

k. Voltage extremes

* 4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first 25 produced and subjected to the following tests as described in paragraph 4.6:

a. Sampling plan A tests

b. High temperature

c. Temperature - altitude

d. Temperature shock

e. Vibration failure

f. Acceleration

g. Humidity

h. Fungus

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i. Salt fog

j. Dust

k. Sealing thermal shock

l. Audio frequency susceptibility

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.5.2.4 Sampling plan C. Unless otherwise specified, 4 indicators shall be selected at random from each 500 or less produced and tested as specified in 4.6.32 for a total of 6,600 hours.

* 4.5.2.4.1 Accept/reject criteria. Figure 3 shall be the criteria used to determine whether the equipment will be accepted or rejected. Certified time records shall be maintained of the test time accumulated on the equipment after starting the environmental cycling specified in 4.6.32. In the event a continue-test decision applies after 6,600 hours of testing, the production quantity shall be considered as conditionally acceptable; however, the accumulated time and failures shall be added to the data of the next production quantity for a composite accept/reject decision. If the continue-test condition persists, the data from the successive production quantities shall be accumulated until 10 failures or 37,540 hours of testing have occurred at which time an accept/reject decision shall be made.

* 4.5.2.4.2 Recording. Recording data handling, and reporting procedures shall be in accordance with MIL-STD-781C.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Turn indicator static balance. The indicator shall be placed in a stationary position, with no power applied, and the pointer checked. Deviation of the pointer from zero shall not exceed 0.010 inch for any stationary position.

4.6.3 Turn indicator dynamic balance. The indicator shall be placed in a stationary position and the pointer checked with the gyro operating on 28V dc. Deviation of the pointer from zero shall not exceed 0.010 inch for any stationary position.

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4.6.4 Dielectric strength. The completely assembled indicator, with filter installed, shall be subjected to a potential of 500V dc applied for a period of 60 seconds each between the positive terminals and conducting parts of the case and between the negative terminals and conducting parts of the case. There shall be no insulation breakdown or other damage as a result of this test.

4.6.5 Turn indicator sensitivity. The indicator shall be placed in the normal position and operating of 28V dc. The indicator shall be turned about the vertical axis at the rates specified in table I. Deflection of the pointer shall be within the tolerances specified therein. This test shall be conducted after power has been applied for a period of not more that 3 minutes. The rate of turn may be measured with a timer, or one indicator may be accurately checked against a timer and used as a standard for checking other indicators. The pointer shall move smoothly throughout its range.

TABLE I. Pointer deflection.

RATE OF TURN (DEGREE PER MINUTE)	DEFLECTION OF POINTER TOLERANCE (INCH)
36	1/16 ± 1/64
90	5/32 ± 1/32
180	5/16 ± 1/32
360	5/8 ± 1/16

4.6.6 Slip indicator sensitivity. With the dial vertical, the indicator shall be rotated to the right about the longitudinal axis until the ball is just short of its limit. The angle of rotation shall be $8^{\circ} \pm 2^{\circ}$. This procedure shall be repeated except the indicator shall be rotated to the left. The same tolerance shall apply. The ball shall not stick at the high end of the tube.

4.6.7 Slip indicator zero position. The two lower adjoining mounting holes of the indicator shall be placed in the normal horizontal position and the indicator shall be gently tapped. The glass ball shall rest at the zero mark of the slip indicator within ± 1/32 inch.

4.6.8 Slip indicator friction. The indicator shall be slowly tipped to either side in the vertical plane of the dial. The indicator shall be gently tapped and the ball shall roll smoothly.

4.6.9 Slip indicator visibility. The indicator shall be tipped so that the ball will be at rest at either end of the tube. When the ball is viewed from a position 12 inches directly in front of the zero mark of the slip indicator, not less than one-half of the ball shall be visible.

4.6.10 Slip indicator filling. The indicator shall be tipped so that all the air in the tubes is trapped in the expansion chamber of the tube. The mounting lugs of the indicator shall be in a vertical plane with the line adjoining the center of the two lower mounting holes in a horizontal plane. When the indicator is viewed from a position 12 inches directly in front of the zero mark of the slip indicator, no air bubbles shall be visible.

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4.6.11 Leakage. The indicator case shall be tested for leaks by means of a mass-spectrometer type of helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.12 Lighting. The lighting circuits of the indicator shall be properly connected and the indicator tested in accordance with MIL-L-25467 or MIL-L-27160, as applicable. Lighting data (brightness and color) shall be taken at the start of reliability testing and at least once every 160 hours.

4.6.12.1 Visual comparison. The prime standard and the indicator under test shall be placed side by side at eye level. The two indicators shall be viewed from a distance of approximately 2 feet in a sighting cone of 30° from the normal of the cover glass surface. Lighting tests on production indicators shall be conducted as follows:

a. Red lighting (MIL-L-25467)

(1) 5.0 $\pm$ 0.1V shall be applied to pins C and D (ground) on the prime standard and the production indicator shall be checked. The two indicators shall be compared and shall appear visually identical in color.

(2) 3.0 $\pm$ 0.1V shall be applied to the prime standard and 3.1V shall be applied to the indicator under test. The production indicator shall appear visually no dimmer than the prime standard.

(3) 3.0 $\pm$ 0.1V shall be applied to the prime standard and 2.9V shall be applied to the indicator being tested. The indicator being tested shall appear visually no brighter than the prime standard.

b. White lighting (MIL-L-27160) - The procedures in 4.6.12.1 shall be repeated.

4.6.12.2 Visual comparison tests. Indicators that do not meet the visual comparison tests specified in 4.6.12.1 shall be subjected to the brightness and color tests of MIL-L-25467 or MIL-L-27160, as applicable, and sufficient data shall be taken to substantiate conformance with the applicable lighting specification. Indicators that also fail these tests shall be rejected. Visual comparison tests shall not be used for first article inspection.

4.6.13 Fogging. The indicator shall be energized in a $71^{\circ} \pm 2^{\circ}$C controlled ambient for a minimum period of 1 hour. At the end of this period and while still at the specified temperature, an ice cube shall be rubbed on the indicator glass face for a period of 2 to 3 minutes. The glass shall be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.14 Turn indicator damping at room temperature. The indicator shall be placed in normal position and operated on 28V dc. The indicator shall be rotated about the vertical axis at such a rate as to cause full-scale deflection of the pointer. When rotation of the indicator is suddenly stopped, the pointer shall return to the zero mark, with no overswing, in not less than 1 second nor more than 3 seconds.

4.6.15 Slip indicator damping at room temperature. The plane of the indicator mounting lugs shall be placed in a vertical position with the line joining the

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centers of the two lower or two upper mounting holes inclined to the horizontal at an angle of 12°. The indicator shall be tipped suddenly through a vertical angle of 24° so that the ball will roll to the opposite end of the tube. The time required for the ball to roll from the zero mark to the end of the tube shall not be less than 0.2 second.

4.6.16 Turn indicator sensitivity at low temperature. The indicator shall be subjected to a temperature of -54° ± 2°C for a period of 4 hours. While still at the specified temperature, the indicator shall be subjected to the test specified in 4.6.5 except that the test shall be conducted after power has been applied for a maximum of 10 minutes. The rate of turn shall be determined with a timer. During this test, the pointer shall deflect 5/16 ± 1/32 inch. When turning is stopped, the pointer shall return smoothly to 0 ± 1/64 inch.

4.6.17 Slip indicator damping at low temperature. The indicator shall be subjected to a temperature of -54° ± 2°C for a period of 4 hours. While still at the specified temperature, the indicator shall be subjected to the test specified in 4.6.15. The time required for the ball to roll from the zero mark to the end of the tube shall not exceed 4 seconds.

4.6.18 Turn indicator sensitivity at high temperature. The indicator shall be subjected to a temperature of 71° ± 2°C for a period of 4 hours. While still at the high temperature, the indicator shall be subjected to the test specified in 4.6.5 except that the test shall be conducted after power has been applied for a maximum period of 10 minutes. The rate of turn shall be determined by use of a timer. The pointer shall deflect 5/16 ± 1/32 inch and shall return smoothly to 0 ± 1/64 inch when the turning is stopped. No failure shall occur during this test.

4.6.19 Slip indicator leakage. With the indicator at room temperature, the size of the air bubbles in the tube shall be determined. The temperature shall be raised to 71° ± 2°C and the indicator maintained at this temperature for a period of not less than 1 hour. The temperature shall then be reduced to room temperature and the size of the air bubbles in the tube again determined. There shall be no appreciable change in the size of the air bubbles before and after the indicator has been subjected to the high temperature. This test may be combined with the test specified in 4.6.18.

4.6.20 Magnetic effort. The indicator, first not operating and then operating on rated voltage, shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 223°, 270°, and 315°, from the initial position. At each of these positions, the indicator shall be rotated on its horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 5°.

4.6.21 Vibration error. The indicator shall be subjected to a vibration test with power applied and the indicator mounted in its normal position. The indicator shall be vibrated with a circular motion in a plane inclined 45° to the horizontal plane at frequencies ranging from 5 to 50 Hz at an amplitude between 0.003 and 0.005 inch. The indicator shall be vibrated from 5 to 50 Hz for 1 minute and from 50 to 5 Hz for 1 minute. Turn indicator oscillation shall not exceed 1/32 inch and slip indicator oscillation shall not exceed 1/16 inch.

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4.6.21.1 Pointer vibration and variation. The maximum amplitude of the pointer vibration and variation shall not exceed 1/32 inch.

4.6.21.2 Turn indicator accuracy. The turn indicator reading shall not differ at any time by more than 1/32 inch from the reading obtained when the instrument is at rest.

4.6.21.3 Slip indicator accuracy. The slip indicator reading shall not differ at any time by more than 1/16 inch from the reading obtained when the instrument is at rest.

4.6.22 Current. A calibrated dc ammeter and voltmeter shall be connected in line between the indicator and the power source. After a 3-minute period of operation, the current required shall not exceed 0.17 amp with an electrical potential of 28V dc at the indicator terminals.

4.6.23 Voltage extremes. With an electrical potential of 26 and 30V dc applied at the indicator terminals, the indicator shall be subjected to and shall pass the sensitivity tests specified in 4.6.5, 4.6.16, and 4.6.18.

* 4.6.24 High temperature. The high temperature exposure test shall be conducted in accordance with MIL-STD-810, method 501.1, procedure I for a period of 24 hours. While still at the high temperature, rated power shall be applied. The indicator shall operate without sticking. After the indicator has returned to room temperature, it shall pass the tests specified in 4.6.5 and 4.6.6. There shall be no damage that would affect subsequent operation.

* 4.6.25 Temperature-altitude. The indicator shall be subjected to a temperature-altitude test in accordance with MIL-STD-810, method 504.1, equipment category 6, procedure I, except in steps 3, 12, and 13, the altitude shall be 80,000 feet, in steps 6 and 8, the temperature shall be 71°C. The indicator shall operate without sticking. After returning to room temperature and atmospheric pressure, the indicator shall meet the individual tests. The indicator shall operate properly throughout this test and there shall be no damage that would affect subsequent operation.

* 4.6.26 Temperature shock. The temperature shock test shall be conducted in accordance with MIL-STD-810, method 503.1, except in step 2, the temperature shall be -54°C (-65°F). After the indicator has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals nor moisture penetration. The indicator shall then be subjected to and shall pass the individual tests.

* 4.6.27 Vibration failure. The indicator shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514.2, equipment category b.1, procedure I, part 1, curve C, except that the upper frequency limit shall be 500 Hz. Power shall be applied for this test. At the end of this test, the indicator shall be subjected to and shall pass the tests specified in 4.6.5 and 4.6.6. No damage to the indicator shall result from this test.

4.6.28 Acceleration. The indicator, not operating, shall be mounted on a centrifuge in its normal operating position and accelerated, first along its vertical axis and then along each of the two axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an

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acceleration of 20g's in each of the axes for 1 minute. At the conclusion of this test, the indicator shall be subjected to and shall meet the individual tests. No damage to the indicator shall result from this test.

4.6.29 Environmental tests. The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810. At the option of the procuring activity, these tests may be performed on finished empty cases with all external parts attached as on production units. At the completion of each test, the case shall be inspected for any damage that would impair indicator operation.

* 4.6.29.1 Humidity. The humidity test shall be conducted in accordance with MIL-STD-810, method 507.1, procedure I, except that in step 1, the temperature shall be 71°C (160°F); in step 4, the relative humidity shall be 95 percent; in step 6, the indicator shall be returned to room ambient before operation, and step 7 shall be completed within 1 hour of completing step 6.

* 4.6.29.2 Fungus.

* 4.6.29.2.1 Purpose. The fungus test is used to determine the resistance of equipment to fungi and to determine if such equipment is adversely affected by fungi under conditions favorable for their development, namely high humidity, warm atmosphere, and presence of inorganic salts.

Typical materials which will support and are damaged by fungi are:

Cotton
Wood
Linen
Cellulose nitrate
Regenerated Cellulose
Leather
Paper and cardboard
Cork
Hair and felts
Natural rubber
Plastic materials containing linen, cotton or wood flour as filler
Vinyl films containing fungus susceptible plasticizers
Formulations of elastomers containing fungus susceptible catalysts, plasticizers or fillers.

* 4.6.29.2.2 Apparatus. The apparatus required to conduct this test consists of chambers or cabinets together with auxiliary instrumentation capable of maintaining the specified condition of temperature and humidity. Provisions shall be made to prevent condensation from dripping on the test item. There shall be free circulation of air around the test item and the surface area of fixtures supporting the test item shall be kept to a minimum.

* 4.6.29.2.3 Procedures.

* 4.6.29.2.3.1 Procedure I.

* 4.6.29.2.3.1.1 Preparation of mineral-salts solution. The solution shall contain the following:

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Potassium dihydrogen orthophosphate (KH_2PO_4)	0.7 g
Potassium monohydrogen orthophosphate (k_2HPO_4)	0.7 g
Magnesium sulfate ($MgSO_4 \cdot 7H_2O$)	0.7 g
Ammonium nitrate (NH_4NO_3)	1.0 g
Sodium chloride (NaCl)	0.005 g
Ferrous sulfate ($FeSO_4 \cdot 7H_2O$)	0.002 g
Zinc sulfate ($ZnSo_4 \cdot 7H_2o$)	0.002 g
Manganous sulfate ($MnSO_4 \cdot 7H_2O$)	0.001 g
Distilled water	1000 ml

Sterilize the mineral salts solution by autoclaving at 121°C (250°F) for 20 minutes. Adjust the pH of the solution by the addition of 0.01 normal solution of NaOH so that after sterilization the pH is between 6.0 and 6.5. Prepare sufficient salts solution for the required tests.

* 4.6.29.2.3.1.1.1 Purity of reagents. Reagent grade chemicals shall be used in all tests. Unless otherwise specified, it is intended that all reagents shall conform to the specifications of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available.

* 4.6.29.2.3.1.1.2 Purity of water. Unless otherwise specified, references to water shall be understood to mean distilled water or water of equal purity.

* 4.6.29.2.3.1.2 Preparation of mixed spore suspension. The following test fungi shall be used:

Fungi	ATCC No.[1]	NLABS No.[2]
Aspergillus niger	9642	386
Aspergillus flavus	9643	380
Aspergillus versicolor	11730	432
Penicillium funiculosum	9644	391
Chaetomium globosum	6205	459

[1]American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852.

[2]Pioneering Research Division, U.S. Army Natick Laboratories, Natick, Massachusetts 01760.

a. Maintain cultures of these fungi separately on an appropriate medium such as potato dextrose agar. However, the culture of Chaetomium globosum shall be cultered on strips of filter paper on the surface of mineral-salts agar. (Mineral salts agar is identical to mineral salts solution described in 4.6.29.2.3.1.1, but contains in addition 15.0 g of agar per liter.) The stock culturers may be kept for not more than

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The spore suspension may be prepared fresh each day or may be held at 6 ± 4°C (43°F) for not more than 4 days.

* 4.6.29.2.3.1.3 Visibility of inoculum control. With each daily group of tests place each of three pieces of sterilized filter paper, 1 inch square, on hardened mineral-salts agar in separate Petri dishes. Inoculate these with the spore suspension by spraying the suspension from a sterilized atomizer (DeVilbiss No. 154 atomizer or equivalent has been found satisfactory for this purpose.) so that the entire surface is moistened with the spore suspension. Incubate these at 29°C (84°F) at a relative humidity not less than 85 percent and examine them after 14 days incubation. There shall be copious growth on all three of the filter paper control specimens. Absence of such growth requires repetition of the test.

* 4.6.29.2.3.1.4 Control items. In addition to the viability of inoculum control, a number of known susceptible substrates shall be inoculated along with the test item to insure that proper conditions are present in the incubation chamber to promote fungus growth. The control items shall include three pieces each of preservative free vegetables tanned leather and protein-glue bonded cork.

* 4.6.29.2.3.1.5 Inoculation of test and control items.

a. Mount the test and control items on suitable fixtures or suspended from hangers.

b. Precondition the chamber and its contents at 29°C (84°F) and 95 percent R.H. for at least 4 hours.

4 months at 6 ± 4°C (43°F). Use subcultures incubated at 29° (84°F) for 7 to 20 days in preparing the spore suspension.

b. Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus a sterile 10-ml portion of water or of a sterile solution containing 0.05 g per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate. Use a sterile platinum or nichrome inoculating wire to scrape gently the surface growth from the culture of the test organism. Pour the spore charge into a sterile 125-ml glass-stoppered Erlenmeyer flask containing 45 ml. of sterile water and 10 to 15 solid glass beads, 5 mm. in diameter. Shake the flask vigorously to liberate the spores from the fruiting bodies and to break the spore clumps.

c. Filter the shaken suspension through a thin layer of sterile glass wool in a glass funnel into a sterile flask in order to remove mycelial fragments.

d. Centrifuge the filtered spore suspension aseptically, and discard the supernatant liquid. Resuspend the residue in 50 ml. of sterile water and centrifuge.

e. Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with sterile mineral salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ± 200,00 spores per ml. as determined with a counting chamber.

f. Repeat this operation for each organism used in the test and blend equal volumes of the resultant spore suspension.

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c. Inoculate the test and control items with the mixed fungus spore suspension (4.6.29.2.3.1.2) by spraying it on the test and control items in the form of a fine mist from a previously sterilized atomizer or nebulizer until they are thoroughly wet with the spray. Incubation is to be started immediately following the inoculation.

* 4.6.29.2.3.1.6 Incubation.

a. Maintain the test chamber at 29°C (84°F) and 95 percent R.H. (minimum) during the life of the test. Keep the test chamber closed during the incubation period except during inspection or for addition of other test items.

b. After 14 days, inspect the control items. They should show an abundant growth of fungus. If the control items do not show an abundant growth, the entire tests shall be repeated.

c. If the control items show satisfactory fungus growth, continue the test for a period of 28 days from the time of inoculation.

* 4.6.29.2.3.1.7 Criteria for passing the test. At the end of the incubation period, the test item shall be removed from the test chamber and inspected in accordance with paragraph 4.6.29.

* 4.6.29.3 Salt fog. The salt fog test shall be conducted in accordance with method 509.1, procedure I for a period of not less than 168 hours.

* 4.6.29.4 Dust. The dust test shall be conducted in accordance with method 510.1 procedure I.

4.6.30 Sealing thermal shock. The indicator case, with mechanisms removed and sealed in the same manner as the complete indicator, shall be immersed alternately in tap water maintained at 85° ± 5°C and 5 ± 4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes with not more than 5 seconds elapsing between bath immersions. No damage to the hermetic seal shall occur as a result of this test. Following the immersions, the indicator case shall be subjected to and shall meet the test specified in 4.6.11.

* 4.6.31 Audio frequency susceptibility. The indicator shall be subjected to and shall meet the audio frequency susceptibility test of MIL-I-6181. During this test, the direct-current consumption shall not change more than 0.020 ampere and pointer deflection shall not change more than 1/64 inch while the audio frequency is being applied.

* 4.6.32 Reliability. The indicators, properly connected and with rated power applied, shall be placed in test chambers which meet the environmental cycle conditions shown on figure 4. During the ON cycle, the indicators shall be exercised by a 3-axis, 30° total movement, 10 cpm, reversing scorsby action and by the vibration shown on figure 4. After each 160-hour cycle of operation, the indicators shall meet the individual tests. Upon completion of this test, or upon reaching the accept decision, the indicators shall pass the sampling plan A tests.

4.6.32.1 Longevity. At the completion of the test specified in 4.6.32, three of the indicators shall continue to be tested for a total of 3,000 hours each.

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* 4.6.32.1.1 Evaluation of longevity test. No accept/reject criteria for this test is specified. The evaluation shall be limited to the analysis of all failures, identification of design and pattern failures, and the submission of recommended design changes to the procuring activity.

4.6.32.2 Relevancy criteria. All failures shall be classified as relevant unless approved for exclusion by the contracting activity.

4.6.32.3 Repair of failed units. In the event of a failure during testing, the failed unit shall be repaired and the test continued.

* 4.6.33 Maintainability demonstration plan. The contractor shall prepare a maintainability demonstration plan in accordance with MIL-STD-471A to determine that the requirements of 3.4.2 have been met.

* 4.7 Inspection and test. Inspection and tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of the specification.

* 5 PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified by the procuring activity.

* 5.1.1 Level A.

* 5.1.1.1 Cleaning. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

* 5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

* 5.1.1.3. Preservation application. Not required.

* 5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with method IC-1 of MIL-P-116. Overbox each completed pack in PPP-B-636 fiberboard, weather-resistant class box. Carton shall be large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the indicators when completed packs are subjected to the rough handling drop tests of MIL-P-116.

* 5.1.2 Level C. Units shall be cleaned, dried and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

* 5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Units packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, styles A or B, class overseas, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and of minimum tare and cube, consistent with the protection required.

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* 5.2.2 Level B. Units packaged as specified in 5.1.1 shall be packed in weather resistant class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above are applicable.

* 5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commerical practice provided the latter meets the requirements of this level.

* 5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ARU-3/A and ARU-19A indicators covered by this specification are intended for use in aircraft to indicate the rates of rotation of the aircraft about its vertical axis and the lateral attitude of the aircraft relative to the apparent vertical.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type and class of indicator required (see 1.2)

c. Whether sampling plans B and C will be conducted (see 4.5.2.2 and 4.5.2.4)

d. Whether a first article is required for inspection and approval (see 3.1 and 4.4)

e. Levels of packaging and packing required (see section 5)

6.3 Definitions.

6.3.1 Tapping. The term "tapped lightly" means the application of light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 Hermetic seal. A hermetic seal is a perfectly closed and airtight seal made between vitric or metallic materials or both. A hermetic seal does not include seals accomplished by gas gaskets.

6.3.3 Amplitude. Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

* 6.3.4 Discrimination ratio. The discrimination ratio is the ratio of the upper-test MTBF (θ_0) to the lower-test MTBF (θ_1). In the event that any conflict in definition exists between this specification and any other document, the definition in this specification will govern.

* 6.4 First article. When a first article is required (see 6.2, d), the item will be inspected and tested in accordance with section 4 of this document. The

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contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

* 6.5 Subject term (key word) listing.

Indicator
Integrally lighted
Turn and slip
2-inch size

* 6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement STANAG 3322 and ASCC Air Std 10/5. When amendment, revision, or cancellation of this specification is proposed which will modify the international standardization agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

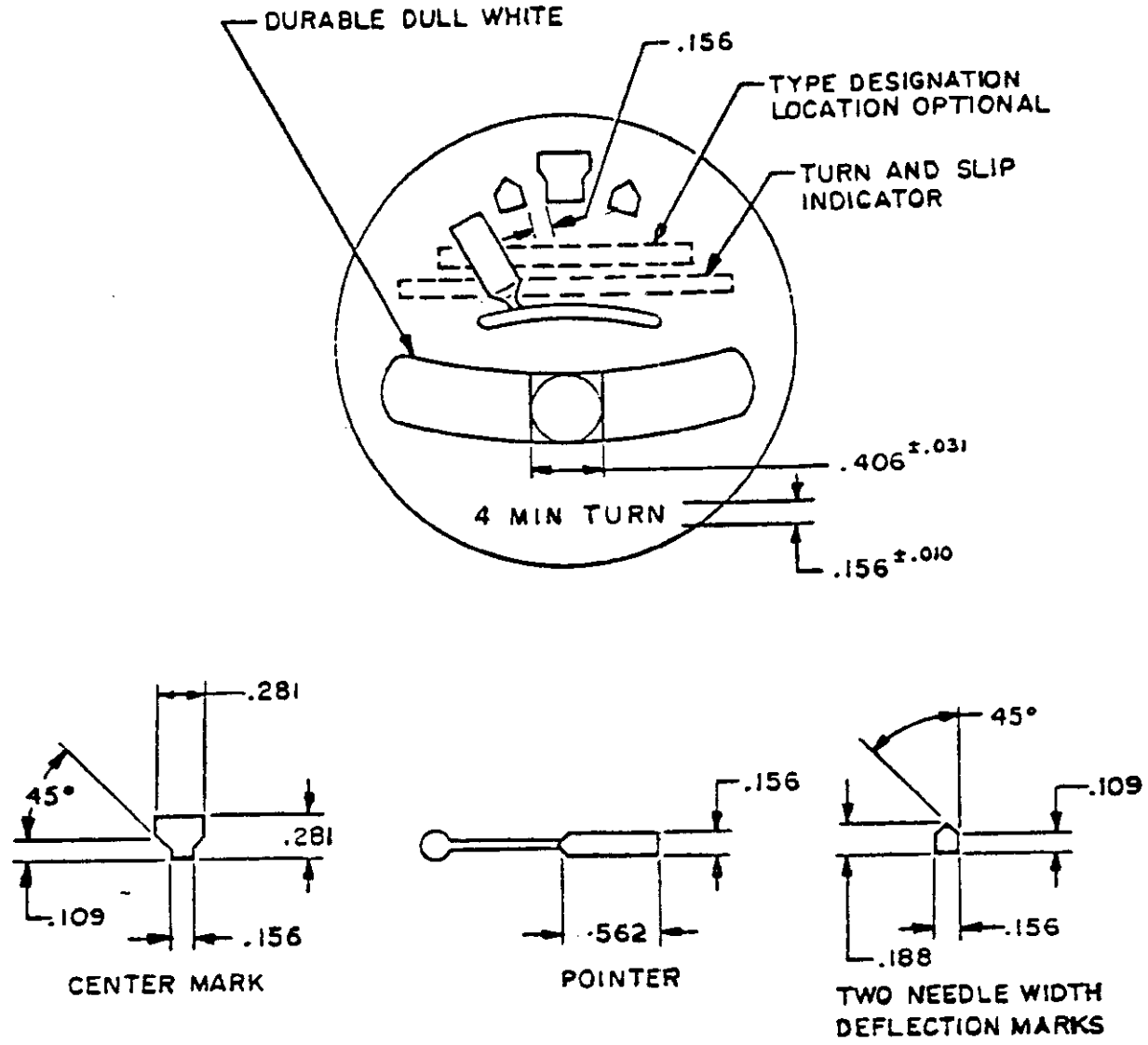
* 6.7 Changes from previous issue. The margins of this revision are marked with an asterick to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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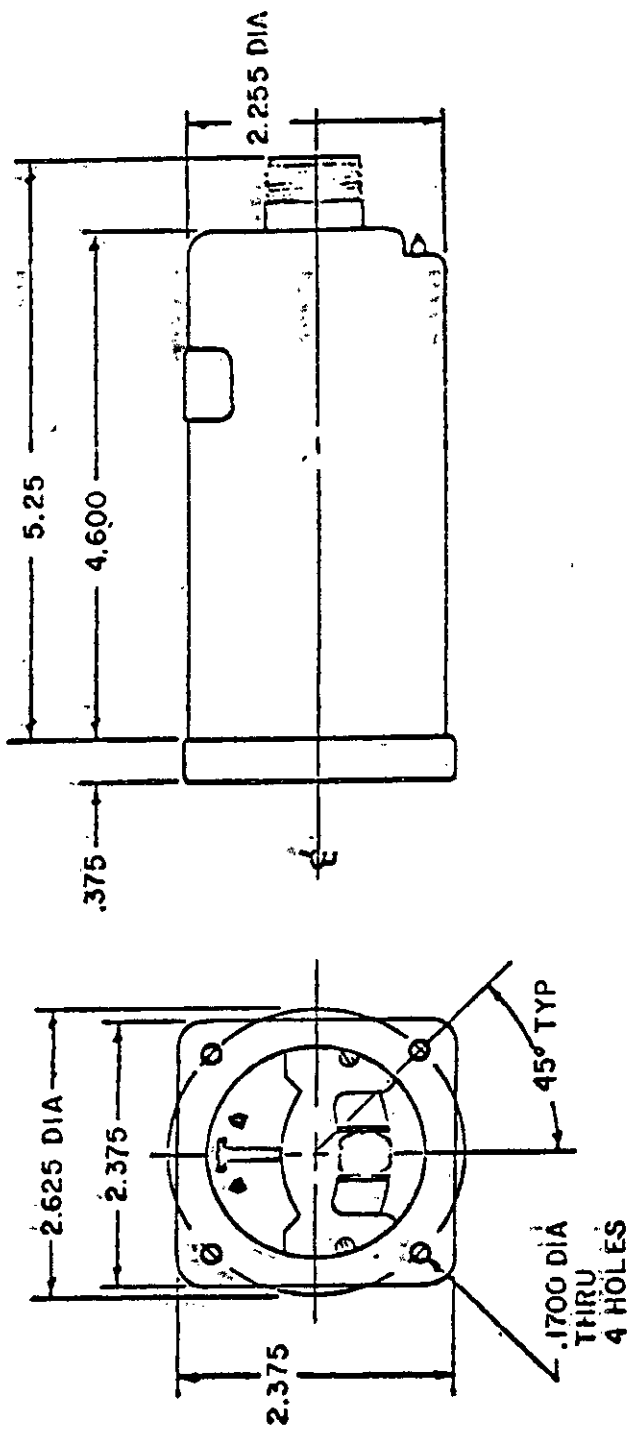
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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010

FIGURE 1. Dial and pointer.

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DIMENSIONS IN INCHES. TOLERANCE: DECIMALS ±.005

FIGURE 2. Case.

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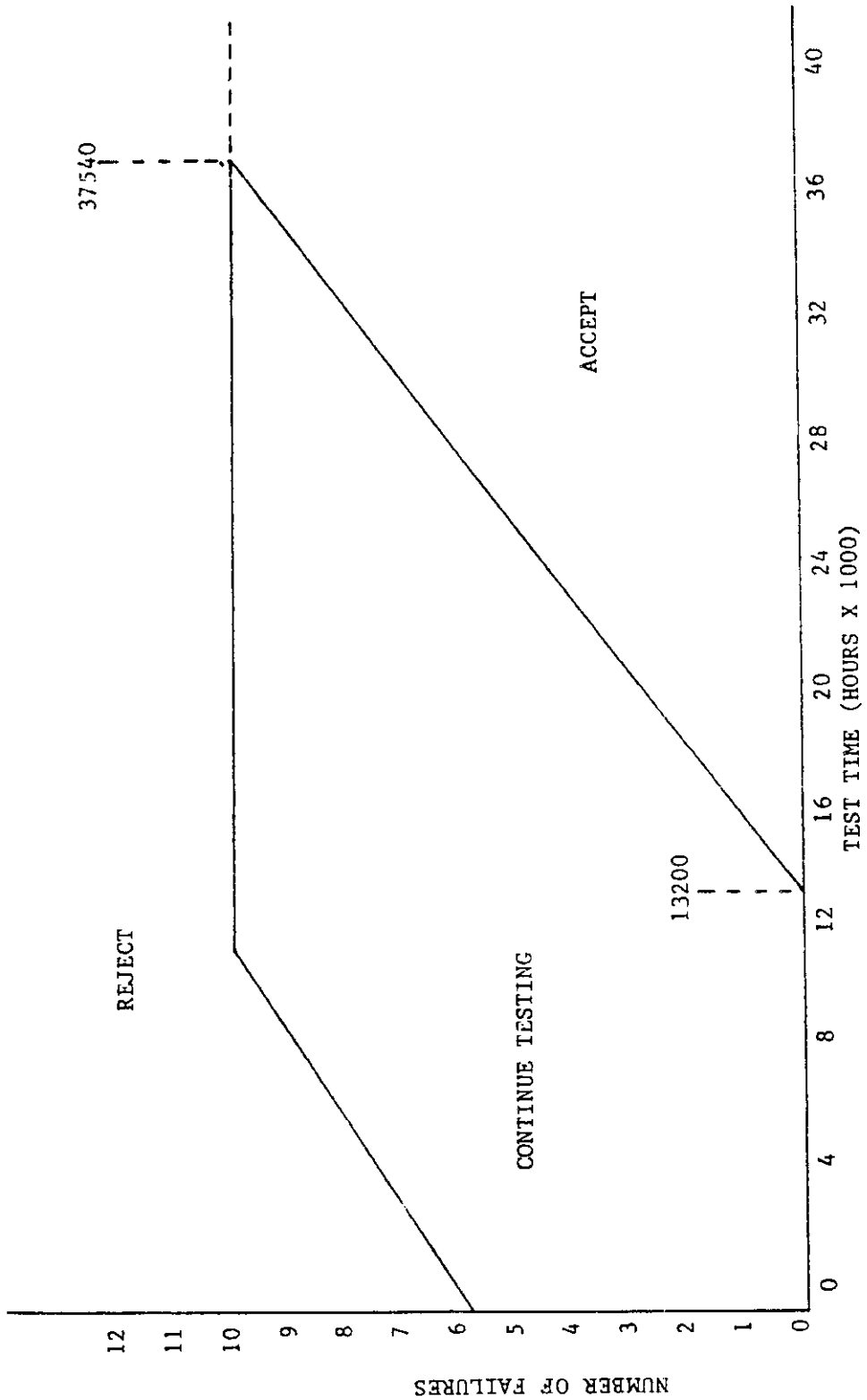
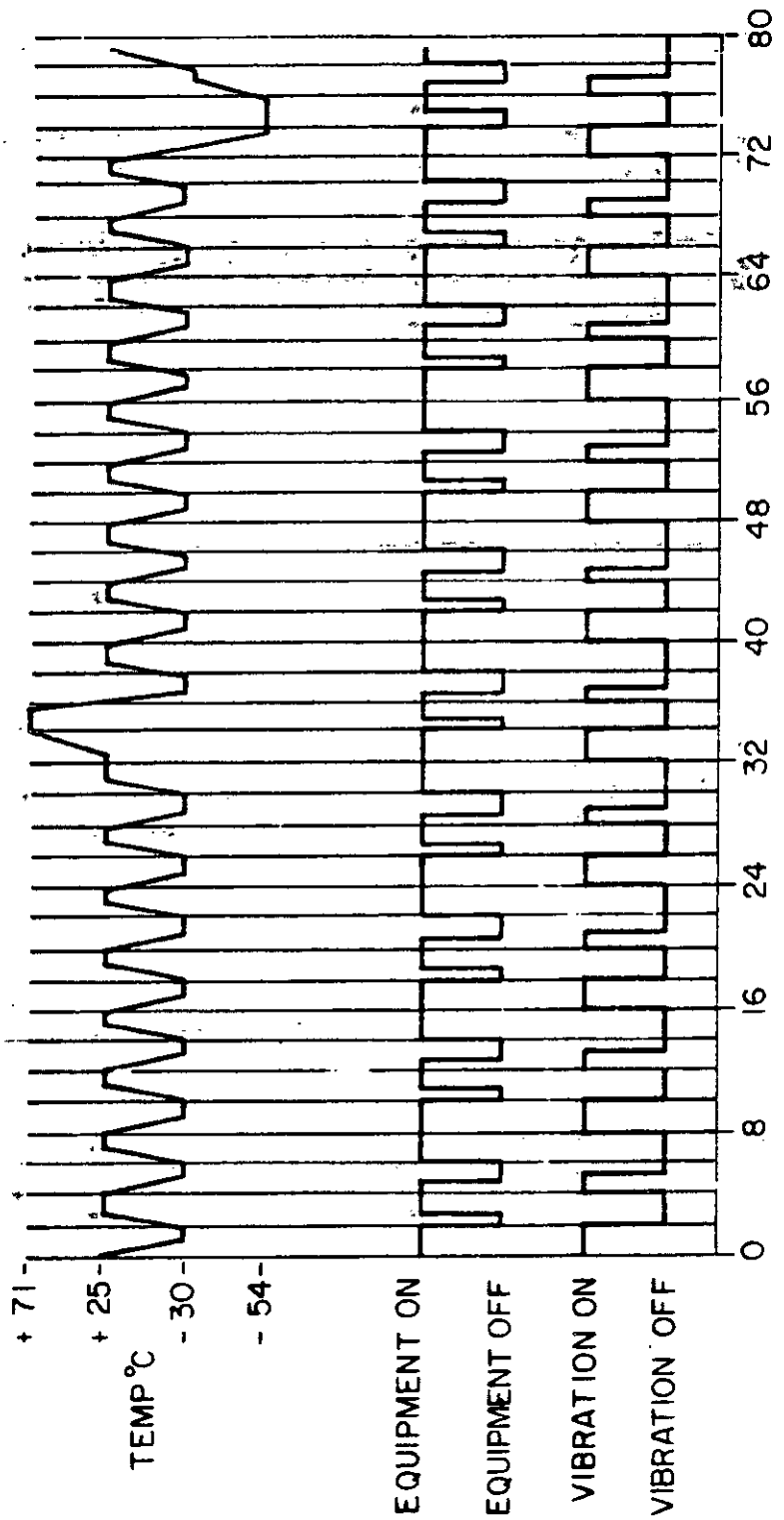


FIGURE 3. Accept-continue test-reject test criteria.

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1) TEMPERATURE CHANGE > 2° C/ MINUTE AVERAGE. PERIODS OF + 71° C, AND - 54° C, TESTS MAY BE CHANGED TO COINCIDE WITH WORKING SHIFT.
2) EQUIPMENT OFF PERIOD = 1/2 HOUR WITHIN THE PERIOD SHOWN.
3) VIBRATION ON PERIOD = 3 HRS/8 HRS APPROX AS SHOWN.
4) TEMPERATURE CHAMBER SHALL BE STABILIZED AT THE TEMPERATURE SHOWN AT EACH HOUR POINT.
5) ENVIRONMENT CYCLE REPEATS AT EACH 80 HOURS
6) VIBRATION LEVEL = 0.5g, 20 TO 30 Hz.
7) MONITORING OF OUTPUTS FOR CATASTROPHIC FAILURE EACH ON CYCLE.
8) MONITORING OF PERFORMANCE CONDUCTED EACH 160 HOURS.
9) ACCUMULATED HOURS - ALL OPERATING TIME.

FIGURE 4. RELIABILITY TEST ENVIRONMENT CYCLES

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MIL-I-25941D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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